HOUFU CHEN

Computational Modeling & Machine Learning Researcher

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PROFILE

Doctoral researcher combining constraint-based systems biology, scientific Python, and machine learning with four engineering internships in automotive and enterprise software. Develops reproducible simulation and validation workflows from SBML metabolic models, literature evidence, and greenhouse measurements. Experienced in explaining technical work clearly through graduate teaching, research documentation, and cross-functional engineering.

CORE SKILLS

Scientific modeling: Flux balance analysis (FBA), dynamic FBA, pFBA, FVA, COBRApy, SBML, linear programming, source-sink modeling, sensitivity analysis, parameter calibration, state estimation

Data and ML: Python, pandas, NumPy, scikit-learn, TensorFlow, PyTorch, PCA, regression, classification, clustering, feature engineering, data visualization

Software and cloud: Git, SQL, Linux/Ubuntu, Docker, AWS Lambda, API Gateway, REST APIs, C/C++, C#, JavaScript

Testing and validation: Selenium, iPerf3, API testing, regression testing, BDD/TDD, SonarQube, network diagnostics, reproducible technical documentation

RESEARCH EXPERIENCE

Doctoral Researcher | University of Waterloo | Waterloo, ON | 2026–Present
Tomato Systems Biology and Greenhouse Digital-Twin Research

- Developing a dynamic flux-balance model of tomato metabolism that couples a VYTOP-derived vegetative source network to a fruit model through nine phloem and xylem transport pools.
- Implemented an hourly Python/COBRApy simulation workflow using SBML models, sequential linear programs, parsimonious FBA, pool-state updates, and development-dependent fruit objectives.
- Evaluated model-to-measurement viability using 45 traceable tomato and leaf mass samples and a 23-day canopy-height series; identified dry mass, fruit age, diameter, and Brix as high-value validation measurements.
- Built evidence and equation audits that distinguish measured, literature-transferred, fitted, and chosen parameters; used carbon-balance, timestep, sensitivity, and boundary-condition checks to expose model limitations before greenhouse calibration.

Research Contributor | Jilin Agricultural Science and Technology University | Jilin, China | 2025
Phosphate Management and Humic-Acid Analysis in Maize Systems

- Analyzed multi-year field-trial data spanning soil chemistry, fluorescence spectroscopy, FTIR, thermogravimetry, and crop yield using Python and principal component analysis.
- Integrated multidimensional measurements into treatment-ranking analyses and supported manuscript structure, statistical interpretation, and technical editing.

TEACHING EXPERIENCE

Teaching Assistant | University of Toronto | Toronto, ON | 2024–2025
APS1070 Foundations of Data Analytics and Machine Learning; CSC108 Introduction to Computer Programming; MIE370 Introduction to Machine Learning

- Led tutorials, office hours, and online support covering Python, data preprocessing, regression, classification, clustering, PCA, neural networks, debugging, and model evaluation.
- Assessed projects and exams for courses serving up to 500+ students per term, applying consistent rubrics and providing actionable feedback on code and methodology.
- Helped students translate mathematical and algorithmic concepts into working experiments while maintaining an inclusive, responsive learning environment.

INDUSTRY EXPERIENCE

Complaints Management and Internal Consulting Intern | FAW-Volkswagen | Changchun, China | May–Jul 2025

- Coordinated high-risk complaint investigations across dealerships and internal teams, evaluating evidence, liability, procedural requirements, and escalation risk.
- Built Kanban dashboards to track case progress, surface recurring issues, and support data-informed process improvement.
- Prepared reasoned case responses and cross-functional summaries for sensitive customer and operational decisions.

WiFi Software Testing Engineer Intern | Ford Motor Company of Canada | Oakville, ON | May–Aug 2022

- Benchmarked SYNC and TCU connectivity with iPerf3 across TCP, UDP, FTP, and HTTP workloads; tested 2.4/5 GHz configurations and multiple security modes.
- Built and maintained Ubuntu client-server test environments and resolved 40+ SSH and Selenium exceptions through dependency, driver, and command-level debugging.
- Authored reusable operating procedures for Ubuntu, iPerf3, SSH, Selenium, and test troubleshooting.

QA Automation Co-op | Manulife, Teranet, and Imagine Communications | Ontario | Jan 2020–Apr 2021

- Developed Selenium-based regression, UI-interaction, and server-monitoring tests in JavaScript and C# across three four-month engineering internships.
- Applied BDD/TDD practices, corrected 40+ SonarQube issues, and improved automated coverage of cross-browser and business-critical workflows.
- Presented defect analysis and a product demonstration to 50+ stakeholders spanning development, business analysis, and product teams.

SELECTED PROJECTS

FoodRacoon | ML-Powered Restaurant Recommender | Python, GPT-4 API, AWS Lambda, API Gateway, Yelp and Google Maps APIs

- Deployed a bilingual, location-aware recommendation service that converts natural-language food preferences into ranked restaurant options.
- Implemented adjustable ranking weights for rating, price, and review count in a serverless application architecture.

Emergency Route Planner | Multi-Agent Optimization Research Project | Python, LLMs, Dijkstra, ant-colony optimization

- Developed an agent-based evacuation-planning framework for testing routing strategies across changing environmental and infrastructure scenarios.
- Used optimization algorithms and LLM-assisted evaluation to compare route decisions and scenario responses.

EDUCATION

University of Waterloo | PhD in Chemical Engineering, in progress
Research focus: constraint-based systems biology, plant metabolism, and greenhouse digital twins

University of Toronto | Master of Engineering, Electrical and Computer Engineering | Sep 2023–Mar 2025
Data Analytics and Machine Learning emphasis | GPA: 3.84/4.00

University of Waterloo | Bachelor of Applied Science, Nanotechnology Engineering, Co-op | Sep 2018–Jun 2023
Graduated with Distinction | President's Scholarship of Distinction | President's Research Award

CREDENTIALS

Machine Learning Specialization | Stanford University and DeepLearning.AI, 2025 | SQL for Data Science | University of California, Davis, 2025 | Algorithms and Data Structures Certificate | Stanford Online, 2025